

Findings Report

To: Quality Management

From: David Mullor Tomas

Re: Lab Quality KPI Analysis Findings Summary

Purpose

This memo summarizes findings from an analysis of lab test and batch release data covering January 2024 to December 2025, conducted to identify quality trends and improvement opportunities across our testing operations.

Headline Findings

1. **Overall batch pass rate stands at 60.5%** across 446 batches and roughly 2,000 individual tests, with a 9.6% test-level fail rate. This leaves meaningful room for improvement in first-pass batch release.
2. **Microbial contamination testing is the leading driver of failures.** Microbial_Count_CFU_g alone accounts for approximately 49% of all recorded test failures nearly half of total failure volume from a single parameter. The next largest contributor, Density_g_ml, adds a further ~16%. Together, these two parameters account for roughly two-thirds of all failures.
3. **Vernier CH is the leading site on test failure.** Example: Vernier CH shows a pass rate of 53.6%, notably below the network average of 60.5%, reaching the lowest point in April 2024 at 16.7%, without a significant decrease in total testing numbers. This warrants closer review of local testing or handling practices.
4. **Fragrance oil shows the highest failure on the secondary cumulative fail% .** Fragrance oil shows a higher failure concentration on Density_g_ml, suggesting a formulation- or process-specific issue rather than a general testing problem.
5. **Monthly pass rate shows volatility rather than a clear trend,** ranging roughly between 40–75% month to month, with May-2025 representing the strongest month and July-2025 the weakest. This suggests failures may be driven more by episodic factors (e.g. specific batches, supplier lots) than a steady underlying decline.

Recommendation

Given that a single test parameter (microbial count) accounts for roughly half of all failures, we recommend a focused review of:

- Sample handling and storage procedures ahead of microbial testing
- Whether current specification limits for microbial count are appropriately calibrated
- Site-level practices at [lowest-performing site], to determine whether the issue is systemic or localized

Methodology Note

This analysis is based on a representative dataset modeled on standard laboratory QC data structures, *(with the format taken from [1] <https://www.nature.com/articles/s41597-022-01203-x>)* the covering test-level results across product lines, sites, and test types. Batch status was determined using a pass / fail / pending logic, where a batch is marked "Pass" only if all associated tests passed and no results are missing.